Reg No: RA2411028020034

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

RAMAPURAM CAMPUS

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMISTRY

B.Tech Degree, CLA I

Sub Code/Name : 21BTB102T/ Computational Biology
Set : B
Class : 1st Year / 1st Sem / B.Tech
Max Marks :50

Date : 18.10.24
Duration: 90 mins

PART A (10 x 1= 10 marks)

ANSWER ALL THE QUESTIONS

Q.No.	Question	Marks	BL	CO	PI
1	Endoplasmic reticulum is absent in _____. (a) Animal Cells (b) Plant cells (c) Prokaryotic cells (d) Eukaryotic cells	1	1	1	4.1.1
2	Crossing over occurs in meiosis during _____. (a) Prophase -1 (b) Prophase -2 (c) Metaphase -1 (d) Metaphase-2	1	1	1	4.1.1
3	Undifferentiated cells are called as_____ . (a) Blood cells (b) Stem cells (c) Epithelial cells (d) Bone marrow cells	1	1	1	4.1.1
4	The correct sequence of different phases of mitosis is __. (a) Anaphase Metaphase Prophase Telophase Interphase (b) Interphase Telophase Metaphase Anaphase Prophase (c) Metaphase Anaphase Telophase Prophase (d) Interphase Prophase Metaphase Anaphase Telophase	1	1	1	4.1.1
5	Cell membrane is _ _____. (a) Unilaminar (b) Bilaminar (c) Trilaminar (d) Tetralaminar	1	1	1	4.1.1
6	Which of the following is the simplest form of carbohydrates? (a) Carboxyl groups (b) Aldehyde and Ketone groups (c) Alcohol and Carboxyl groups (d) Hydroxyl groups and Hydrogen groups	1	1	2	1.2.1
7	Which of these is NOT a lipid? (a) Fats (b) Oils (c) Proteins (d) Waxes	1	1	2	1.2.1

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8	Hormones which regulates blood sugar are____ and _____. (a) Glycogen and Glucagon (b) Insulin and Glucagon (c) Glucose and Glycogen (d) Glucagon and Pituitary	1	1	2	1.2.1
9	Sugar in RNA is_____. (a) Ribose (b) Deoxy ribose (c) Hexose (d) Triose	1	1	2	1.2.1
10	DNA sequencing followed by genome annotation are steps of _____. (a) Comparative genomics (b) Structural genomics (c) Functional genomics (d) Transcriptomics	1	1	2	1.2.1
	Part – B 3 x 8 marks = 24 marks **Answer ANY 3 Questions**				
11	i) What are the differences between Plant cell and Animal cell? ii) Explain the structure and functions of chloroplast.	4	1	1	4.1.1
12	i) Write about Embryonic Stem Cells. ii) Write short note on Interphase (Cell Cycle).	4 4	1	1	4.1.1
13	i) Write about nucleic acid and its components (DNA &RNA). ii) Write a note on structural and functional genomics.	4 4	1	2	1.2.1
14	Briefly explain about Human Genome Project (HGP).	8	2	2	1.2.1
	Part – C 16 marks **Answer ANY 1 Question**				
15	Give the ethical and legal issues of using stem cell.	16	2	1	4.2.1
16	Justify why structure of a protein determines the function of the protein.	16	2	2	4.1.1

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